

DSG-CTTA Project Formula Handbook & Mathematical Reference Guide

Project Title: Disparity-Aware Continual Test-Time Adaptation for Accent-Robust Automatic Speech Recognition
Extended Title: Disparity-Aware Continual Test-Time Adaptation for Accent-Robust ASR: Characterizing and Controlling Adaptation-Induced Performance Disparities
Governing Standard: Architectural Decision Record 005 (ADR-005), Stage 0 Protocol Freeze, Stage 5D/5E Locked Protocol
Document Status: FROZEN REFERENCE SPECIFICATION & PEDAGOGICAL HANDBOOK

Executive Overview: How to Read This Handbook

This document serves as the authoritative mathematical and operational reference for all equations used throughout the DSG-CTTA research program. For each equation, this handbook provides:

- The Mathematical Expression:** Standard formal formulation.
- Where the Formula Was Obtained (Literature Provenance):** Exact primary literature source, paper title, authors, venue, year, DOI/URL, and original equation/section number.
- What Each Parameter Represents in the Dataset:** Concrete mapping to specific CSV filenames, column headers, speaker clusters, clip counts, and reference word sums.
- Real Project Values & Step-by-Step Arithmetic:** Concrete numbers from the frozen experimental suite plugged directly into the equation.
- Units and Valid Ranges:** Theoretical bounds versus empirical ranges observed in our experiments.
- Operational Interpretation:** Clear explanations of what higher and lower values mean for speech recognition accuracy and demographic fairness.
- Source Code Implementation:** Exact repository file paths and function names.

1. Word Error Rate (WER)

1.1 Mathematical Formulation

$$\text{WER} = \frac{S + D + I}{N_{\text{ref}}}$$

where: - S = Count of word substitutions - D = Count of word deletions - I = Count of word insertions - N_{ref} = Total count of reference words in ground truth

1.2 Where the Formula Was Obtained (Literature Provenance)

- Primary Source:** Levenshtein, V. I. (1966). *Binary codes capable of correcting deletions, insertions, and reversals*. Soviet Physics Doklady, 10(8), 707–710.
- ASR Standardization:** Hunt, M. J. (1990). *Figures of merit for assessing connected-word recognisers*. Speech Communication, 9(4), 329–336. Formalized by the National Institute of Standards and Technology (NIST) Speech Recognition Benchmark Protocol (1990–present).
- Official Documentation:** NIST Scoring Toolkit (SCTK) / `sclite` mathematical specification.

1.3 What Each Parameter Represents in the Dataset

- **Evaluation Partition:** `datasets/splits/stage5_external_eval.csv` (Holdout stream) and `datasets/splits/stage5_sentinel_panel.csv` (Sentinel panel).
- **S (Substitutions):** The count of words in the ground-truth transcript (`sentence` column) that were incorrectly replaced by a different vocabulary word in the model's decoded hypothesis. Computed via dynamic programming alignment over tokenized strings.
- **D (Deletions):** The count of words present in the ground-truth transcript (`sentence` column) that the model completely failed to transcribe (omitted words).
- **I (Insertions):** The count of extraneous words produced in the model's hypothesis that do not correspond to any spoken token in the reference text.
- **N_{ref} (Total Reference Words):** The exact total word count of all normalized sentences in the target evaluation split. In `stage5_external_eval.csv` ($N = 900$ clips across 60 speakers), $N_{\text{ref}} = 8667$ words.
- US English (150 clips): $N_{\text{us}} = 1468$ words (16.94%)
- England English (150 clips): $N_{\text{england}} = 1393$ words (16.07%)
- South Asian English (150 clips): $N_{\text{south_asian}} = 1430$ words (16.50%)
- Australian English (150 clips): $N_{\text{australia}} = 1493$ words (17.23%)
- Canadian English (150 clips): $N_{\text{canada}} = 1500$ words (17.31%)
- Irish English (150 clips): $N_{\text{ireland}} = 1383$ words (15.96%)
- Exact Sum Invariant: $1468 + 1393 + 1430 + 1493 + 1500 + 1383 = 8667$ words.

1.4 Real Project Values & Step-by-Step Arithmetic

Evaluated on `facebook/wav2vec2-base-960h` across the 900-clip holdout stream (`reports/stage5/final_external_metrics.csv`): - **No-Adapt Baseline (θ_0):**

$$\text{WER}(\theta_0) = \frac{1583 + 178 + 185}{8667} = \frac{1946}{8667} \approx 0.2245298 \implies 22.45\%$$

- **SUTA Unconstrained Adaptation (θ_{SUTA}):**

$$\text{WER}(\theta_{\text{SUTA}}) = \frac{1645 + 240 + 156}{8667} = \frac{2041}{8667} \approx 0.2354909 \implies 23.55\%$$

- **DSG Controlled Adaptation (θ_{DSG}):**

$$\text{WER}(\theta_{\text{DSG}}) = \frac{1590 + 181 + 181}{8667} = \frac{1952}{8667} \approx 0.2252221 \implies 22.52\%$$

1.5 Units, Valid Ranges, and Interpretation

- **Unit:** Unitless ratio, reported as a percentage (%) by multiplying by 100.
- **Theoretical Range:** $[0, +\infty)$. Bounded below by 0.0 (perfect transcription). Can exceed 1.0 (100%) when insertions outnumber reference words ($I > N_{\text{ref}}$).
- **Project Observed Range:** 9.18% (`distil_whisper_small` No-Adapt on CV27) to 96.38% (`whisper_tiny` on L2-ARCTIC Stage 2).

- **Interpretation:** Lower is better. WER = 0% indicates exact bit-for-bit match. Higher values indicate increasing transcription errors.

1.6 Source Code Implementation

- **File:** `src/dsg_ctta/reporting/metrics.py`
- **Function:** `compute_wer(reference: str, hypothesis: str) -> Dict[str, Any]`

2. Character Error Rate (CER)

2.1 Mathematical Formulation

$$\text{CER} = \frac{S_c + D_c + I_c}{N_{\text{char}}}$$

where: - S_c = Count of character substitutions - D_c = Count of character deletions - I_c = Count of character insertions - N_{char} = Total character count in reference text

2.2 Where the Formula Was Obtained (Literature Provenance)

- **Primary Source:** Standard phonetic transcription and speech recognition metric, widely adopted for morphologically rich or character-based languages and sub-word phonetic analysis.
- **Citation:** Morris, A. C., Maier, V., & Green, P. (2004). *From WER and RIL to MER and WIL: improved evaluation measures for connected speech recognition*. Interspeech 2004.

2.3 What Each Parameter Represents in the Dataset

- S_c (**Character Substitutions**): Individual characters incorrectly substituted in the model prediction relative to normalized characters in `sentence`.
- D_c (**Character Deletions**): Characters in `sentence` omitted by the model.
- I_c (**Character Insertions**): Spurious characters emitted by the model.
- N_{char} : Total characters in the reference text (excluding stripped punctuation).

2.4 Real Project Values & Step-by-Step Arithmetic

On `wav2vec2_base` (`reports/stage5/final_external_metrics.csv`): - Baseline No-Adapt: $\text{CER}(\theta_0) = 0.102234 \implies 10.22\%$ - SUTA Adaptation: $\text{CER}(\theta_{\text{SUTA}}) = 0.104866 \implies 10.49\%$ - DSG Adaptation: $\text{CER}(\theta_{\text{DSG}}) = 0.095476 \implies 9.55\%$ (Acoustic character precision improved under DSG).

2.5 Units and Operational Interpretation

- **Unit:** Percentage (%).
- **Theoretical Range:** $[0, +\infty)$.
- **Project Observed Range:** 9.55% (DSG on CV27) to 77.08% (`whisper_tiny` on L2-ARCTIC).
- **Interpretation:** Measures sub-word acoustic transcription precision. Lower values indicate higher phonetic fidelity.

3. Disparity Range (D)

3.1 Mathematical Formulation

$$D = \max_{g \in \mathcal{G}} \text{WER}_g - \min_{g \in \mathcal{G}} \text{WER}_g$$

3.2 Where the Formula Was Obtained (Literature Provenance)

- **Primary Source:** Rai, S., et al. (2025). *Measuring and Benchmarking Equity Across Speech Recognition Systems (ASR-FAIRBENCH)*. Proceedings of Interspeech 2025, Section 3.1 ("Performance Disparity Metrics").
- **Foundational Context:** Koenecke, A., et al. (2020). *Racial disparities in automated speech recognition*. Proceedings of the National Academy of Sciences (PNAS), 117(14), 7684–7689.
- **Formal Adoption in Project:** Formalized in Stage 0 Protocol Freeze ([docs/protocol_freeze.md](#)) and ADR-005 ([docs/adr/ADR-005-stage5-gate-rule.md](#)).

3.3 What Each Parameter Represents in the Dataset

- $\mathcal{G}$ (**Set of Evaluated Strata**): The 6 pre-specified accent groups in [stage5_external_eval.csv](#), derived from self-reported metadata in Mozilla Common Voice 27.0 via the deterministic grouping rule ([protocol_amendment_cv27.md](#)):

$$\mathcal{G} = \{\text{US, England, South Asian, Australian, Canadian, Irish}\}$$

- WER_g : The aggregate Word Error Rate evaluated strictly over the 150 clips belonging to accent group g .
- $\max_{g \in \mathcal{G}} \text{WER}_g$: The highest error rate observed among the six accent strata (consistently **South Asian English** across all models).
- $\min_{g \in \mathcal{G}} \text{WER}_g$: The lowest error rate observed among the six accent strata (typically **Canadian English** or **Irish English**).

3.4 Real Project Values & Step-by-Step Arithmetic

On [facebook/wav2vec2-base-960h](#) ([reports/stage6/eight_model_group_metrics.csv](#)): - **No-Adapt Baseline** (θ_0):

$$\max_{g \in \mathcal{G}} \text{WER}_g(\theta_0) = \text{WER}_{\text{South Asian}}(\theta_0) = \frac{607}{1430} \approx 42.4476\%$$

$$\min_{g \in \mathcal{G}} \text{WER}_g(\theta_0) = \text{WER}_{\text{Canadian}}(\theta_0) = \frac{183}{1500} \approx 12.2000\%$$

$$D(\theta_0) = 42.4476\% - 12.2000\% = 30.2476 \text{ pp} \approx 30.25 \text{ pp}$$

- **DSG Controlled Adaptation** (θ_{DSG}):

$$\max_{g \in \mathcal{G}} \text{WER}_g(\theta_{\text{DSG}}) = \text{WER}_{\text{South Asian}}(\theta_{\text{DSG}}) = \frac{603}{1430} \approx 42.1678\%$$

$$\min_{g \in \mathcal{G}} \text{WER}_g(\theta_{\text{DSG}}) = \text{WER}_{\text{Canadian}}(\theta_{\text{DSG}}) = \frac{190}{1500} \approx 12.6667\%$$

$$D(\theta_{\text{DSG}}) = 42.1678\% - 12.6667\% = 29.5011 \text{ pp} \approx 29.50 \text{ pp}$$

- **Disparity Reduction under DSG:**

$$\Delta D = D(\theta_{\text{DSG}}) - D(\theta_0) = 29.5011 \text{ pp} - 30.2476 \text{ pp} = -0.7465 \text{ pp} \approx -0.75 \text{ pp}$$

3.5 Units, Valid Ranges, and Interpretation

- **Unit:** Percentage points (pp).
- **Theoretical Range:** $[0, +\infty)$ (typically $[0, 100 \text{ pp}]$ when group WERs are bounded within 100%).
- **Project Observed Range:** 8.92 pp (`xlsr_english` No-Adapt) to 47.83 pp (`whisper_tiny` Stage 2).
- **Interpretation:** Measures performance inequality. A value of $D = 0 \text{ pp}$ represents perfect demographic parity (all groups experience identical error rates). Higher values indicate severe equity disparities.

3.6 Source Code Implementation

- **File:** `src/dsg_ctta/reporting/metrics.py`
- **Function:** `compute_disparity(group_wers: Dict[str, float]) -> float`

4. Disparity Ratio (R)

4.1 Mathematical Formulation

$$R = \frac{\max_{g \in \mathcal{G}} \text{WER}_g}{\min_{g \in \mathcal{G}} \text{WER}_g}$$

4.2 Where the Formula Was Obtained (Literature Provenance)

- **Primary Source:** Algorithmic fairness and civil rights compliance literature (Four-Fifths Rule / Adverse Impact Ratio). Formalized for ASR by Rai et al. (Interspeech 2025, ASR-FAIRBENCH).

4.3 Real Project Values & Step-by-Step Arithmetic

On `wav2vec2_base` No-Adapt:

$$R(\theta_0) = \frac{42.4476\%}{12.2000\%} \approx 3.4793 \implies 3.48\times$$

Interpretation: Speakers from South Asia experienced 3.48 **times more transcription errors** than speakers from Canada on the unadapted baseline model.

5. Overall Regression (Δ_R)

5.1 Mathematical Formulation

$$\Delta_R(\theta') = \text{WER}(\theta') - \text{WER}(\theta_{\text{ref}})$$

5.2 Where the Formula Was Obtained (Literature Provenance)

- **Primary Source:** ADR-005: Stage 5 Disparity Safety Gate Decision Rule and Rollback Semantics ([docs/adr/ADR-005-stage5-gate-rule.md](#)), Section 3. Adapted from continual learning catastrophic forgetting metrics (Kirkpatrick et al., PNAS 2017).

5.3 What Each Parameter Represents in the Dataset

- θ' (**Candidate Model**): The in-memory shadow model state produced by taking unsupervised entropy minimization gradient steps on the current unlabeled batch B_t ($K = 4$ audio clips from [stage5_external_eval.csv](#)).
- θ_{ref} (**Reference Model**): In online gating, this is the current validated live model θ_t . In post-hoc reporting, this is the unadapted baseline model θ_0 .
- $\text{WER}(\theta)$: The aggregate Word Error Rate evaluated across the 300 clips of the independent sentinel panel ([datasets/splits/stage5_sentinel_panel.csv](#)).

5.4 Real Project Values & Step-by-Step Arithmetic

Evaluated on [facebook/wav2vec2-base-960h](#) holdout stream ([reports/stage5/final_external_metrics.csv](#)): - **SUTA Overall Regression vs Baseline** ($\theta_{\text{ref}} = \theta_0$):

$$\Delta_R(\theta_{\text{SUTA}}) = \text{WER}(\theta_{\text{SUTA}}) - \text{WER}(\theta_0)$$

$$\Delta_R(\theta_{\text{SUTA}}) = 23.5491\% - 22.4530\% = +1.0961 \text{ pp} \approx +1.10 \text{ pp}$$

- **DSG Overall Regression vs Baseline** ($\theta_{\text{ref}} = \theta_0$):

$$\Delta_R(\theta_{\text{DSG}}) = \text{WER}(\theta_{\text{DSG}}) - \text{WER}(\theta_0)$$

$$\Delta_R(\theta_{\text{DSG}}) = 22.5222\% - 22.4530\% = +0.0692 \text{ pp} \approx +0.07 \text{ pp}$$

- **Net Relative Improvement (DSG vs SUTA):**

$$\Delta_{\text{DSG vs SUTA}} = \text{WER}(\theta_{\text{DSG}}) - \text{WER}(\theta_{\text{SUTA}})$$

$$\Delta_{\text{DSG vs SUTA}} = 22.5222\% - 23.5491\% = -1.0269 \text{ pp} \approx -1.03 \text{ pp}$$

5.5 Units, Valid Ranges, and Interpretation

- **Unit:** Percentage points (pp).
- **Theoretical Range:** $(-\infty, +\infty)$.
- **Project Observed Range:** -0.12 pp (`data2vec_base` DSG) to $+2.73 \text{ pp}$ (`xlsr_english` SUTA).
- **Interpretation:** $\Delta_R > 0$ indicates aggregate performance degradation (harm). $\Delta_R \leq 0$ indicates neutral or improved performance.

6. Subgroup Regression (Δ_g) & Maximum Subgroup Regression ($\max_g \Delta_g$)

6.1 Mathematical Formulation

$$\Delta_g(\theta') = \text{WER}_g(\theta') - \text{WER}_g(\theta_{\text{ref}})$$

$$\max_{g \in \mathcal{G}} \Delta_g(\theta') = \max_{g \in \mathcal{G}} (\text{WER}_g(\theta') - \text{WER}_g(\theta_{\text{ref}}))$$

6.2 Where the Formula Was Obtained (Literature Provenance)

- **Primary Source:** ADR-005 (`docs/adr/ADR-005-stage5-gate-rule.md`), Section 3. Derived from minimax group robustness and distributionally robust optimization (Sagawa et al., ICLR 2020, *Distributionally Robust Neural Networks*).

6.3 What Each Parameter Represents in the Dataset

- Δ_g : The error rate difference on a specific stratum $g \in \mathcal{G}$ (e.g. `Irish English`, containing 150 clips in holdout, 50 clips in sentinel).
- $\max_g \Delta_g$: The single largest error rate inflation suffered by any of the 6 accent strata, identifying the most severely harmed demographic subgroup.

6.4 Real Project Values & Step-by-Step Arithmetic

On `facebook/wav2vec2-base-960h` under SUTA:

$$\Delta_{\text{US}} = 21.32\% - 21.46\% = -0.14 \text{ pp} \quad (\text{Improved})$$

$$\Delta_{\text{England}} = 21.82\% - 20.32\% = +1.50 \text{ pp} \quad (\text{Regressed})$$

$$\Delta_{\text{South Asian}} = 42.66\% - 42.45\% = +0.21 \text{ pp} \quad (\text{Regressed})$$

$$\Delta_{\text{Australia}} = 24.05\% - 22.91\% = +1.14 \text{ pp} \quad (\text{Regressed})$$

$$\Delta_{\text{Canada}} = 13.73\% - 12.20\% = +1.53 \text{ pp} \quad (\text{Regressed})$$

$$\Delta_{\text{Ireland}} = 18.00\% - 15.62\% = +2.38 \text{ pp} \quad (\text{Severe Regression!})$$

$$\max_{g \in \mathcal{G}} \Delta_g(\theta_{\text{SUTA}}) = \max(-0.14, +1.50, +0.21, +1.14, +1.53, +2.38) = +2.38 \text{ pp} \approx +2.39 \text{ pp}$$

Under DSG on the same model:

$$\max_{g \in \mathcal{G}} \Delta_g(\theta_{\text{DSG}}) = +0.47 \text{ pp} \quad (\text{Canadian English: } 12.20\% \rightarrow 12.67\%)$$

Result: DSG constrained the worst-case subgroup regression from +2.39 pp down to +0.47 pp.

7. Disparity Change (ΔD)

7.1 Mathematical Formulation

$$\Delta D(\theta') = D(\theta') - D(\theta_{\text{ref}})$$

7.2 Where the Formula Was Obtained (Literature Provenance)

- **Primary Source:** ADR-005 ([docs/adr/ADR-005-stage5-gate-rule.md](#)), Section 3.

7.3 Real Project Values & Step-by-Step Arithmetic

- Under SUTA on `wav2vec2_base`:

$$\Delta D(\theta_{\text{SUTA}}) = 28.92 \text{ pp} - 30.25 \text{ pp} = -1.33 \text{ pp} \approx -1.32 \text{ pp}$$

- Under DSG on `wav2vec2_base`:

$$\Delta D(\theta_{\text{DSG}}) = 29.50 \text{ pp} - 30.25 \text{ pp} = -0.75 \text{ pp}$$

8. Paired Speaker-Cluster Bootstrap & Upper Confidence Bound (UCB)

8.1 Mathematical Formulation

Given sentinel panel $\mathcal{S}_{\text{sentinel}}$ partitioned into $C = 30$ independent speaker clusters $\mathcal{C} = \{c_1, c_2, \dots, c_{30}\}$: 1. For replicate $b = 1, 2, \dots, B$ ($B = 1000$): - Draw $C = 30$ speaker clusters with replacement:

$$\mathcal{C}^{(b)} = \{c_{i_1}^{(b)}, c_{i_2}^{(b)}, \dots, c_{i_{30}}^{(b)}\}$$

- Aggregate paired errors and words across all clips belonging to sampled clusters.
- Compute replicate metric:

$$\Delta^{(b)} = \text{Metric}(\theta'; \mathcal{C}^{(b)}) - \text{Metric}(\theta_{\text{ref}}; \mathcal{C}^{(b)})$$

1. Compute the empirical 95th percentile Upper Confidence Bound:

$$\text{UCB}_{95}(\Delta) = \text{Quantile}_{0.95} \left(\{\Delta^{(1)}, \Delta^{(2)}, \dots, \Delta^{(B)}\} \right)$$

8.2 Where the Formula Was Obtained (Literature Provenance)

- **Foundational Theory:** Efron, B., & Tibshirani, R. J. (1993). *An Introduction to the Bootstrap*. Chapman & Hall / CRC.
- **Cluster Bootstrapping:** Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2008). *Bootstrap-based improvements for inference with clustered errors*. The Review of Economics and Statistics, 90(3), 414–427.
- **ASR Cluster Application:** Formalized by SPEAKABLE Workshop (LREC 2026) to account for intra-speaker acoustic correlation, operationalized for online CTTA in ADR-005.

8.3 What Each Parameter Represents in the Dataset

- **Clusters** $c_k \in \mathcal{C}$: The 30 unique speaker identifiers in column `client_id` of `datasets/splits/stage5_sentinel_panel.csv`. Exactly 5 unique speakers per accent group, each contributing exactly 10 audio clips.
- **Resampling Unit:** Entire speaker clusters are resampled. If speaker `cmu_spk_042` is drawn twice, all 10 of their utterances are included twice, preserving intra-speaker acoustic covariance.
- $B = 1000$: Number of pseudo-replicate draws, locked for numerical stability.
- $\alpha = 0.05$ (**95% Confidence**): Significance level ensuring risk-averse upper bounds.

8.4 Source Code Implementation

- **File:** [src/dsg_ctta/controller/bootstrap.py](#)
- **Function:** `compute_speaker_cluster_bootstrap_stats(...)`

9. Disparity Safety Gate (DSG) Acceptance Contract

9.1 Mathematical Formulation

$$\text{ACCEPT}(\theta') \iff [\text{UCB}_{95}(\Delta_R(\mathcal{S}_{\text{sentinel}})) \leq \epsilon_R] \wedge \left[\text{UCB}_{95} \left(\max_{g \in \mathcal{G}} \Delta_g(\mathcal{S}_{\text{sentinel}}) \right) \leq \epsilon_G \right] \wedge [\text{UCB}_{95}(\Delta_D(\mathcal{S}_{\text{sentinel}})) \leq \epsilon_D]$$

$$\text{REJECT}(\theta') \iff \neg \text{ACCEPT}(\theta')$$

9.2 Where the Formula Was Obtained (Literature Provenance)

- **Primary Source:** Authoritative Architectural Decision Record 005 ([ADR-005](#)), approved September 30, 2026.
- **Pre-Registration:** Configured in [configs/stage5_gate_config.json](#) (SHA-256: `18faea44...`).

9.3 Frozen Operating Risk Tolerances (ϵ)

- $\epsilon_R = 0.0000$ (0.00 pp): Zero tolerance for overall regression. Candidate must have 95% statistical confidence of zero aggregate degradation on the sentinel panel.
- $\epsilon_G = 0.0200$ (2.00 pp): Subgroup regression tolerance, locked to match calibration threshold δ_G .
- $\epsilon_D = 0.0200$ (2.00 pp): Disparity growth tolerance, locked to match calibration threshold δ_D .

9.4 Real Project Decisions across 1,350 Candidate Evaluations

- **Accepted Updates: 23 updates** (1.70%) across 6 CTC backbones.
- **Rejected Updates: 1,327 updates** (98.30%) via statistical hypothesis testing.
- **Fail-Closed Software Errors: 0 errors.**

10. Percentage Added Error Reduction

10.1 Mathematical Formulation

$$\text{Reduction}_{\text{errors}} = \frac{\Delta \text{Errors}(\theta_{\text{SUTA}}) - \Delta \text{Errors}(\theta_{\text{DSG}})}{\Delta \text{Errors}(\theta_{\text{SUTA}})} \times 100\%$$

where:

$$\Delta\text{Errors}(\theta) = \text{Errors}(\theta) - \text{Errors}(\theta_0)$$

10.2 Where the Formula Was Obtained (Literature Provenance)

- **Primary Source:** Standard intervention efficacy metric in statistical epidemiology and ML safety benchmarks.

10.3 Real Project Values & Step-by-Step Arithmetic

On [facebook/wav2vec2-base-960h](#) across the 8,667-word holdout stream: - Baseline Errors (θ_0): $\text{Errors}(\theta_0) = 1946$ words - SUTA Errors (θ_{SUTA}): $\text{Errors}(\theta_{\text{SUTA}}) = 2041$ words $\implies \Delta\text{Errors}(\theta_{\text{SUTA}}) = 2041 - 1946 = 95$ errors - DSG Errors (θ_{DSG}): $\text{Errors}(\theta_{\text{DSG}}) = 1952$ words $\implies \Delta\text{Errors}(\theta_{\text{DSG}}) = 1952 - 1946 = 6$ errors - Prevented Errors: $95 - 6 = 89$ word errors prevented

$$\text{Reduction}_{\text{errors}} = \frac{95 - 6}{95} \times 100\% = \frac{89}{95} \times 100\% \approx 93.6842\% \implies 93.68\%$$

Interpretation: The Disparity Safety Gate prevented 93.68% of the added transcription errors that occurred under unconstrained SUTA.

11. Confounder-Controlled Poisson GLMM Count Regression

11.1 Mathematical Formulation

$$\log(\mu_{ij}) = \beta_0 + \sum_{g=2}^G \beta_g \mathbb{I}(\text{group}_{ij} = g) + \beta_{\text{SNR}} Z_{\text{SNR},ij} + \beta_{\text{rate}} Z_{\text{rate},ij} + u_{\text{speaker}(i)} + \log(N_{ij})$$

$$\text{Rate Ratio (RR)}_g = \exp(\beta_g)$$

11.2 Where the Formula Was Obtained (Literature Provenance)

- **Foundational Theory:** McCullagh, P., & Nelder, J. A. (1989). *Generalized Linear Models* (2nd ed.). Chapman & Hall / CRC.
- **ASR Fairness Application:** SPEAKABLE Workshop (LREC-COLING 2026), *Responsible Benchmarking of Fairness for Automatic Speech Recognition*.
- **Stage 2 Verification:** Executed and documented in [reports/stage2_final_report.md](#), Section 8.

11.3 What Each Parameter Represents in the Dataset

- $\mu_{ij} = \mathbb{E}[Y_{ij}]$: The expected count of word errors ($S + D + I$) for utterance j spoken by speaker i .
- $\log(N_{ij})$ (**Offset**): Natural logarithm of reference words in [sentence](#). Entering $\log(N)$ with a fixed regression slope of 1.0 converts the count model into an error rate model (μ/N).
- $\mathbb{I}(\text{group}_{ij} = g)$: Dummy variable indicating speech variety stratum g relative to the reference stratum ([Arabic](#)).
- $Z_{\text{SNR},ij}$: Standardized acoustic Signal-to-Noise Ratio (z -score), computed from the raw WAV file via WADA-SNR.

- $Z_{\text{rate},ij}$: Standardized speech articulation tempo (z -score), measured in spoken words per minute.
- $u_{\text{speaker}(i)} \sim \mathcal{N}(0, \sigma_u^2)$: Random intercept accounting for speaker clustering.

11.4 Real Project Values & Step-by-Step Arithmetic

On `wav2vec2_base` in Stage 2 (60 utterances, 6 speaker clusters): - Raw Group Disparity Rate Ratio: $1.183\times$ - Confounder-Adjusted Rate Ratio: $1.124\times$ ($p < 0.05$ with Holm-Bonferroni correction). *Finding*: Disparity across accent groups remains statistically significant even after statistically controlling for acoustic SNR and speech rate.

12. Unsupervised Test-Time Adaptation Loss (SUTA)

12.1 Mathematical Formulation

$$\mathcal{L}_{\text{SUTA}}(\mathbf{x}; \theta) = \mathcal{L}_{\text{ent}}(\mathbf{x}; \theta) + \lambda_{\text{mcc}} \mathcal{L}_{\text{mcc}}(\mathbf{x}; \theta)$$

$$\mathcal{L}_{\text{ent}}(\mathbf{x}; \theta) = -\frac{1}{T'} \sum_{t=1}^{T'} \sum_{v \in \mathcal{V}} P(v \mid \mathbf{x}_t; \theta) \log P(v \mid \mathbf{x}_t; \theta)$$

$$\mathcal{L}_{\text{mcc}}(\mathbf{x}; \theta) = \sum_{i=1}^{|\mathcal{V}|} \sum_{j \neq i} \hat{C}_{ij}^2, \quad \text{where } \hat{\mathbf{C}} = \frac{\tilde{\mathbf{Y}}^\top \tilde{\mathbf{Y}}}{\mathbf{1}^\top \tilde{\mathbf{Y}}^\top \tilde{\mathbf{Y}} \mathbf{1}}$$

12.2 Where the Formula Was Obtained (Literature Provenance)

- **Primary Source**: Lin, G.-T., Lai, C.-I., Chiang, W.-L., & Lee, H.-y. (2022). *Listen, Adapt, Better WER: Source-free Single-utterance Test-time Adaptation for ASR*. Interspeech 2022, Equations (1), (2), and (3).
- **Minimum Class Confusion Theory**: Jin, Y., et al. (ECCV 2020), *Minimum Class Confusion for Versatile Domain Adaptation*.

12.3 What Each Parameter Represents in the Dataset

- $\mathbf{x}$: Streaming audio window B_t ($K = 4$ audio clips from `stage5_external_eval.csv`, ~15 seconds total audio).
 - $P(v \mid \mathbf{x}_t; \theta)$: The CTC frame emission probability over vocabulary $\mathcal{V}$ ($|\mathcal{V}| = 32$) output by the model's linear projection at 20 ms frame t .
 - $\lambda_{\text{mcc}} = 0.1$: Pre-registered hyperparameter weighting class confusion minimization.
 - **Updated Parameters**: Gradient updates are restricted strictly to LayerNorm affine parameters (γ, β).
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Summary Master Table: Equations, Provenance, and Dataset Parameters

Equation #	Metric / Concept	Mathematical Expression	Literature Source & Provenance	Exact Dataset Parameter Mapping	Real Project Value
1	Word Error Rate (WER)	$WER = \frac{S+D+I}{N_{ref}}$	Levenshtein (1966); NIST ASR Standard (1990)	<code>stage5_external_eval.csv</code> , <code>sentence</code> column ($N_{ref} = 8667$ words)	22.45% (No-Adapt) 23.55% (SUTA) 22.52% (DSG)
2	Character Error Rate (CER)	$CER = \frac{S_c+D_c+I_c}{N_{char}}$	Morris et al. (Interspeech 2004)	Sub-word character edit operations against normalized <code>sentence</code> text	10.22% (No-Adapt) 9.55% (DSG)
3	Disparity Range (D)	$D = \max_{g \in \mathcal{G}} WER_g - \min_{g \in \mathcal{G}} WER_g$	Rai et al. (Interspeech 2025, ASR-FAIRBENCH)	6 strata in <code>stage5_accent_group</code> (150 clips, ~1445 words/stratum)	30.25 pp (No-Adapt) 29.50 pp (DSG)
4	Disparity Ratio (R)	$R = \frac{\max_{g \in \mathcal{G}} WER_g}{\min_{g \in \mathcal{G}} WER_g}$	Four-Fifths Adverse Impact Rule; Rai et al. (2025)	Ratio of worst stratum (South Asian) to best stratum (Canadian)	3.48× (No-Adapt)
5	Overall Regression (Δ_R)	$\Delta_R = WER(\theta') - WER(\theta_{ref})$	ADR-005 Section 3; Kirkpatrick et al. (2017)	Evaluated over 300 clips (2925 words) of <code>stage5_sentinel_panel.csv</code>	+1.10 pp (SUTA) +0.07 pp (DSG)
6	Max Subgroup Regression	$\max_{g \in \mathcal{G}} \Delta_g = \max_{g \in \mathcal{G}} (WER'_g - WER_g^{ref})$	ADR-005 Section 3; Sagawa et al. (ICLR 2020)	Maximum localized regression across 6 strata in sentinel panel	+2.39 pp (SUTA) +0.47 pp (DSG)
7	Disparity Change (ΔD)	$\Delta D = D(\theta') - D(\theta_{ref})$	ADR-005 Section 3	Expansion or contraction of cross-accent disparity spread	−1.32 pp (SUTA) −0.75 pp (DSG)
8	Cluster Bootstrap UCB	$UCB_{95} = \text{Quantile}_{0.95}(\{\Delta^{(b)}\})$	Efron & Tibshirani (1993); Cameron et al. (2008)	Resamples 30 speaker clusters in <code>client_id</code> of sentinel panel ($B = 1000$)	Replicate bounds for gating
9	DSG Acceptance Rule	$UCB(\Delta_R) \leq \epsilon_R \wedge UCB(\max_g \Delta_g) \leq \epsilon_G \wedge UCB(\Delta_D) \leq \epsilon_D$	ADR-005 (ADR-005-stage5-gate-rule.md)	Frozen tolerances: $\epsilon_R = 0.0000$, $\epsilon_G = 0.0200$, $\epsilon_D = 0.0200$	23 Accepted (1.7%) 1327 Rejected (98.3%)
10	Added Error Reduction	$\frac{\Delta \text{Error}_{SUTA} - \Delta \text{Error}_{DSG}}{\Delta \text{Error}_{SUTA}} \times 100\%$	ML Intervention Efficacy Standard	Added error difference over 8667 holdout words (95 SUTA vs 6 DSG)	93.68% errors prevented
11	Poisson GLMM	$\log(\mu) = \beta_0 + \sum \beta_g G_g + \beta_z Z + u_{spk} + \log(N)$	McCullagh & Nelder (1989);	Discrete error counts with $\log(\text{words})$ offset, SNR, and speech	Adjusted $RR = 1.124\times$

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	Regression		SPEAKABLE (2026)	rate	($p < 0.05$)
12	SUTA Loss	$\mathcal{L}_{\text{SUTA}} = \mathcal{L}_{\text{ent}} + \lambda_{\text{mcc}} \mathcal{L}_{\text{mcc}}$	Lin et al. (Interspeech 2022, SUTA Eq. 1–3)	Unlabeled audio frames in streaming window B_t ($K = 4$ clips)	$\lambda_{\text{mcc}} = 0.1$